

FILLMORE COUNTY AP, MN, USA

WMO: 720283

Lat: 43.677N

Lon: 92.180W

Elev: 1276

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04927

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-6.3	-17.7	2.2	-9.9	-13.5	2.8	-5.9	28.6	20.5	25.9	16.7	8.8	310	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	87.5	73.4	84.3	71.2	81.7	69.7	76.0	83.3	74.1	81.5	72.4	79.4	11.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	131.3	80.4	71.9	123.5	79.2	70.0	115.7	77.4	40.3	83.3	38.5	82.0	36.8	79.3	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	21.1	19.0	DB	-17.8	91.9	5.1	3.2	-21.4	94.2	-24.4	96.1	-27.3	97.9	-30.9	100.2	(4)
(5)			WB	-18.2	79.2	5.0	2.5	-21.8	81.0	-24.7	82.5	-27.5	83.9	-31.1	85.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	16.5	18.7	32.9	46.0	57.9	67.4	70.8	68.1	62.3	49.0	35.7	22.0
	DBStd	21.49	12.79	12.01	12.20	10.01	8.90	5.79	5.32	5.54	7.62	9.34	10.66	11.19
	HDD50	4127	1040	875	547	190	29	0	0	0	5	133	440	868
	HDD65	7556	1505	1295	996	571	253	39	12	30	140	504	878	1333
	CDD50	2579	0	0	16	71	275	522	646	561	374	102	12	0
	CDD65	532	0	0	1	1	34	111	193	126	58	8	0	0
	CDH74	3922	0	0	4	28	306	807	1463	843	408	63	0	0
	CDH80	938	0	0	0	4	82	191	386	180	86	9	0	0
Wind	WSAvg	9.0	10.4	10.1	10.0	10.6	9.5	8.0	6.8	6.3	7.7	9.2	10.1	9.7
Precipitation	PrecAvg	35.5	0.9	1.0	1.9	3.7	4.2	5.9	4.5	4.8	3.4	2.2	1.8	1.2
	PrecMax	44.0	2.7	2.6	4.1	7.4	8.6	11.4	11.6	14.0	9.5	6.9	6.3	2.8
	PrecMin	23.7	0.3	0.1	0.0	0.6	1.7	2.5	0.9	1.1	1.1	0.4	0.1	0.2
	PrecStd	5.7	0.6	0.6	1.0	2.0	1.7	2.7	2.2	2.9	2.2	1.6	1.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	49.5	72.4	77.7	88.0	90.0	92.5	88.4	86.5	80.9	67.1	50.2
		MCWB	38.8	45.0	59.5	59.3	69.6	73.2	75.8	75.4	70.8	64.5	55.6	46.2
	2%	DB	38.1	42.4	62.2	71.7	81.5	85.2	87.9	84.4	82.3	73.8	60.4	42.7
		MCWB	34.8	38.9	54.6	56.7	64.7	69.9	75.2	73.2	69.5	61.1	51.7	40.0
	5%	DB	35.2	37.7	55.4	66.4	76.9	81.6	84.3	81.7	79.0	69.3	54.8	39.1
		MCWB	33.2	34.9	49.2	53.5	62.4	67.8	72.4	71.8	67.3	59.8	48.1	36.9
	10%	DB	32.4	34.3	49.4	62.5	72.7	78.8	81.5	78.8	75.3	64.0	50.5	36.1
		MCWB	31.2	32.1	44.3	51.3	61.3	66.7	70.7	69.6	65.2	55.5	44.6	34.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	45.6	61.6	61.7	72.5	75.8	80.1	78.2	75.2	68.8	57.4	46.5
		MCDB	42.8	49.4	71.7	72.2	82.8	85.2	87.5	84.9	84.0	74.6	64.1	48.6
	2%	WB	35.3	39.1	56.5	58.8	68.4	73.4	76.8	75.4	71.9	64.6	53.2	40.6
		MCDB	37.7	42.1	61.1	68.2	76.9	81.4	83.7	81.2	79.2	71.7	58.4	42.2
	5%	WB	33.2	35.2	49.2	55.6	65.5	71.1	74.5	73.5	69.4	60.5	48.6	36.9
		MCDB	34.7	37.5	55.4	64.0	73.8	78.2	82.3	79.6	76.0	66.6	54.4	39.0
	10%	WB	31.5	32.3	44.2	52.5	62.5	69.0	72.6	71.4	67.2	56.7	45.3	34.1
		MCDB	32.3	33.9	49.5	61.0	69.6	76.4	80.1	77.0	74.1	64.0	49.5	35.6
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.3	14.6	17.4	18.0	17.3	17.5	17.1	18.9	16.9	14.2	12.3
		MCDBR	14.7	15.8	22.7	25.2	23.0	20.7	20.0	19.1	21.5	23.6	20.6	14.7
		MCWBR	12.8	13.0	14.4	14.2	11.2	10.0	9.7	10.1	10.9	12.9	13.8	12.4
	5% WB	MCDBR	14.1	15.4	21.1	21.9	19.3	17.9	18.1	17.3	18.6	19.0	19.2	14.1
		MCWBR	12.6	13.0	14.1	13.8	11.0	10.7	10.3	10.1	11.1	13.2	13.8	12.3
Clear-Sky Solar Irradiance	taub	0.264	0.293	0.317	0.362	0.390	0.400	0.414	0.399	0.368	0.323	0.289	0.262	
	taud	2.401	2.319	2.391	2.315	2.297	2.301	2.261	2.316	2.376	2.490	2.505	2.447	
	Ebn at Noon	279	286	293	286	280	276	271	271	272	273	268	269	
	Edh at Noon	24	31	33	39	41	41	42	38	33	26	22	21	
All-Sky Solar Radiation	RadAvg	561	894	1196	1448	1710	1905	1975	1676	1346	867	572	420	
	RadStd	58	94	128	144	130	144	89	114	98	119	49	38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page